CSC3105

Data Analytics Sample Quiz 1 Solution

Sample Quiz 1 - Trimester 2, AY2025/26

xx, xxth Feb 2026

x.xxam – x.xxam

(2 hours)

Instructions to students:

1. This Quiz comprises **FOUR(4)** Questions printed on **Fifteen (15)** pages including the cover page.
2. You must answer **ALL** the questions.
3. All answers must be written in this answer sheet.

At the end of the examination,

Please write your **Student ID** on each answer book used.

Failure to do so will mean that your work will not be identified.

The University reserves the right not to mark your script if you fail to follow these instructions.

Answer all the questions (Total 100 marks)

Question 1 (20 marks)

Google search engine deal with more than 3.5 billion searches per day. The main reason for quick query response is that Google uses Term Frequency-Inverse Document Frequency (TF-IDF) approach behind the scene to match each query in millions of documents and websites using proximity measures. It will match those websites and documents with the query and score them according to the proximity measure used. Hence, it will rank and return those that are highly matched in descending order accordingly.

Assume that Google has computed millions of website and documents with the following TF-IDF scores for the following websites that contains following wordings as followed.

Website 1 : Data Analytics is not about statistical calculus mathematics and algorithms but physical insight and visualization on data with decision making
Website 2: Data Analytics seems daunting with mathematics and algorithms tools but is fun decision tools
Website 3: Data Analytics is linked to data decision making tools with data visualization

For these 3 websites, the TF-IDF scores for the wordings "Data" , "Analytics" are as follows:

	Website 1	Website 2	Website 3
Data	0.0307	0.0219	0.0921
Analytics	0.05	0.0714	0.1

For the user query on the word Data Analytics , the TF-IDF score is as follows:

	TF-IDF
Data	0.1534
Analytics	0.5

Perform the Cosine similarity proximity measure on TF-IDF of the user query and the three websites. Hence, find the website that the Google engine will rank the highest that match the user query and return the response back to user based on cosine similarity.

Similarly use the Euclidean distance proximity measure on TF-IDF of the user query and the three websites, find the website that the Google will rank the highest and return the response back to user based on Euclidean distance.

Comparing the two proximity measures, should Google adopt Cosine similarity or Euclidean distance and state the reason with their vector diagram sketch.

Hint: $\cos(\mathbf{x},\mathbf{y}) = \frac{<\mathbf{x},\mathbf{y}>}{\|\mathbf{x}\|\|\mathbf{y}\|}$ where $\mathbf{x}$ and $\mathbf{y}$ are the query and respective website accordingly.
$\mathrm{d}(\mathbf{x},\mathbf{y}) = \|\mathbf{x}-\mathbf{y}\|$

Solution :

cosine similarity proximity measure

For website1

$$\cos(query, website1) = \frac{< query, website1 >}{\|query\|\|website1\|}$$

where

$< query, website1 >$=0.1534*0.0307+0.5*0.05=0.0297

$$\|query\| = \sqrt{(0.1534)^2 + (0.5)^2} = 0.5230$$

$$\|website1\| = \sqrt{(0.0307)^2 + (0.05)^2} = 0.0587$$

Therefore

$$\cos(query, website1) = \frac{0.0297}{0.5230 * 0.0587} = 0.9674$$

For website2

$$\cos(query, website2) = \frac{< query, website2 >}{\|query\|\|website2\|}$$

where

$$< query, website2 >= 0.1534 * 0.0219 + 0.5 * 0.0714 = 0.039$$

$$\|query\| = \sqrt{(0.1534)^2 + (0.5)^2} = 0.5230$$

$$\|website2\| = \sqrt{(0.0219)^2 + (0.0714)^2} = 0.0747$$

Therefore

$$\cos(query, website2) = \frac{0.039}{0.5230 * 0.0747} = 0.9983$$

For website3

$$\cos(query, website3) = \frac{< query, website3 >}{\|query\|\|website3\|}$$

where

$$< query, website3 >= 0.1534 * 0.0921 + 0.5 * 0.1 = 0.0641$$

$$\|query\| = \sqrt{(0.1534)^2 + (0.5)^2} = 0.5230$$

$$\|website3\| = \sqrt{(0.0921)^2 + (0.1)^2} = 0.1359$$

Therefore

$$\cos(query, website3) = \frac{0.0641}{0.5230 * 0.1359} = 0.9019$$

Since the cosine similarity for website 2 is the higher, Google will rank and return the website 2 as the first response to user query on Data Analytics

<u>Euclidean Distance proximity measure</u>

$$\mathrm{d}(query, website1) = \sqrt{(0.1534 - 0.0307)^2 + (0.5 - 0.05)^2} = 0.4664$$
$$\mathrm{d}(query, website2) = \sqrt{(0.1534 - 0.0219)^2 + (0.5 - 0.0714)^2} = 0.4483$$
$$\mathrm{d}(query, website3) = \sqrt{(0.1534 - 0.0921)^2 + (0.5 - 0.1)^2} = 0.4047$$

Since the Euclidean distance for website 3 is the smallest, Google will rank and return the website 3 as the first response to user query on Data Analytics

Question 2 (30 marks)

Given a data matrix $\mathbf{x} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ where column represents the two features namely number of Mac user and number of Window users respectively. The row represents the two data record. Derive out the PC components of this data matrix x. Obtain the dimension reduction using just 1 PC component.

Hint: $\text{cov}(\mathbf{x}) = \frac{1}{N-1}(\mathbf{x} - \boldsymbol{\mu})^T(\mathbf{x} - \boldsymbol{\mu})$

<u>Solution :</u>

Data matrix $\mathbf{x} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$$\text{cov}(\mathbf{x}) = \frac{1}{N-1}(\mathbf{x} - \boldsymbol{\mu})^T(\mathbf{x} - \boldsymbol{\mu})$$

where $\boldsymbol{\mu}$ is the mean of the data for each column and N is the number of data record.

$$\boldsymbol{\mu} = \begin{bmatrix} \frac{2+0}{2} & \frac{0+2}{2} \\ \frac{2+0}{2} & \frac{0+2}{2} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Therefore

$$\text{cov}(\mathbf{x}) = \frac{1}{2-1}\left(\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\right)^T \left(\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\right)$$

$$\text{cov}(\mathbf{x}) = \left(\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\right)^T \left(\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\right) = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1*1 + (-1*-1) & 1*-1 + (-1*1) \\ -1*1 + (1*-1) & -1*-1 + (1*1) \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

Step 2: Find eigenvalues and eigenvectors

Let λ be the eigenvalues of the cov($\mathbf{x}$)

$$\det(\lambda I - \text{cov}(\mathbf{x})) = 0$$

For λI-cov(X)

$$\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} \lambda-2 & 2 \\ 2 & \lambda-2 \end{bmatrix}$$

Therefore

$$\det(\lambda I\text{-cov}(\mathbf{x}))=0$$
$$\left|\begin{bmatrix}\lambda\text{-}2 & 2\\ 2 & \lambda\text{-}2\end{bmatrix}\right|=0$$
$$(\lambda\text{-}2)(\lambda\text{-}2)-2*2=0$$
$$\lambda^2-2\lambda-2\lambda+4-4=0$$
$$\lambda^2-4\lambda=0$$
$$\lambda(\lambda-4)=0\rightarrow\lambda=0\ or\ \lambda=4$$

For $\lambda=0$

$$(\lambda I\text{-cov}(\mathbf{x}))\begin{bmatrix}v_1\\ v_2\end{bmatrix}=0$$
$$\begin{bmatrix}\lambda\text{-}2 & 2\\ 2 & \lambda\text{-}2\end{bmatrix}\begin{bmatrix}v_1\\ v_2\end{bmatrix}=0$$
$$\begin{bmatrix}0\text{-}2 & 2\\ 2 & 0\text{-}2\end{bmatrix}\begin{bmatrix}v_1\\ v_2\end{bmatrix}=0$$
$$\begin{bmatrix}\text{-}2 & 2\\ 2 & \text{-}2\end{bmatrix}\begin{bmatrix}v_1\\ v_2\end{bmatrix}=0$$

Using first row

$$-2v_1+2v_2=0$$
$$\rightarrow\quad v_1=v_2$$

Therefore first eigenvector is

$$\mathbf{v_1}=\begin{bmatrix}v_1\\ v_2\end{bmatrix}=\begin{bmatrix}1\\ 1\end{bmatrix}$$

if $v_2=1$

For $\lambda=4$

$$(\lambda I\text{-cov}(X))\begin{bmatrix}v_1\\ v_2\end{bmatrix}=0$$

$$\begin{bmatrix}\lambda\text{-}2 & 2\\ 2 & \lambda\text{-}2\end{bmatrix}\begin{bmatrix}v_1\\ v_2\end{bmatrix}=0$$
$$\begin{bmatrix}4\text{-}2 & 2\\ 2 & 4\text{-}2\end{bmatrix}\begin{bmatrix}v_1\\ v_2\end{bmatrix}=0$$
$$\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}\begin{bmatrix}v_1\\ v_2\end{bmatrix}=0$$

Using first row

$$2v_1+2v_2=0$$
$$\rightarrow\quad v_1=-v_2$$

Therefore second eigenvector is

$$\mathbf{v_2}=\begin{bmatrix}v_1\\ v_2\end{bmatrix}=\begin{bmatrix}-1\\ 1\end{bmatrix}$$

if $v_2=1$

Since second eigenvector has the eigenvalue of 4 which is the largest power, therefore the second eigenvector will be used as the primary eigenvector principal component

The projection of PC1 on data will be given as

$$(\mathbf{x}-\boldsymbol{\mu}).\mathbf{v_2} = \left(\begin{bmatrix}2 & 0\\0 & 2\end{bmatrix} - \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}\right).\begin{bmatrix}-1\\1\end{bmatrix}$$
$$= \begin{bmatrix}1 & -1\\-1 & 1\end{bmatrix}\begin{bmatrix}-1\\1\end{bmatrix}$$
$$= \begin{bmatrix}1*-1+-1*1\\-1*-1+1*1\end{bmatrix} = \begin{bmatrix}-2\\2\end{bmatrix}$$

Hence 2x2 matrix of $\mathbf{x}$ that contains 2 features become 2x1 vectors that contains one PC component that capture the 2 features. Hence, dimension reduction!!!

Question 3 (30 marks)

Given the table below, find out which features should be the best split to be the main first node.

Name	Can Fly	Live In Water	Class
Human	No	No	Mammal
Python	No	No	Non- Mammal
Salmon	No	Yes	Non- Mammal
Whale	No	Yes	Mammal
Komodo	No	No	Non- Mammal
Bat	Yes	No	Mammal
Pigeon	Yes	No	Non- Mammal
Cat	No	No	Mammal
Leopard Shark	No	Yes	Non- Mammal
Porcupine	No	No	Mammal
Eel	No	Yes	Non- Mammal
Gila Monster	No	No	Non- Mammal
Platypus	No	No	Mammal
Owl	Yes	No	Non- Mammal
Dolphin	No	Yes	Mammal
Eagle	Yes	No	Non- Mammal

Hint : $Entropy = -\sum_{i=1}^{N} p_i \, log_2(pi)$ where N is the number of classes

<u>Solution :</u>

Parent (root)	Count	Probability of Class
Class 1 : Mammals	7	$p_1 = \frac{7}{16}$
Class 2 : Non Mammals	9	$p_2 = \frac{9}{16}$
Entropy=-$\sum_{i=1}^{2} p_i \, log_2(pi) = -p_1 log_2 p_1 - p_2 log_2 p_2 = -\frac{7}{16} log_2 \frac{7}{16} - \frac{9}{16} log_2 \frac{9}{16}$ =0.989		

Live in Water	N1 (Yes= 1)	N1 Probability of Class	N2 (No= 0)	N2 Probability of Class
Class 1: Mammals	2	$p_1 = \frac{2}{5}$	5	$p_1 = \frac{5}{11}$
Class 2: Non-Mammals	3	$p_2 = \frac{3}{5}$	6	$p_2 = \frac{6}{11}$
$Entropy$		$-\frac{2}{5}log_2\frac{2}{5} - \frac{3}{5}log_2\frac{3}{5} = 0.970$		$-\frac{5}{11}log_2\frac{5}{11} - \frac{6}{11}log_2\frac{6}{11} = 0.99$
$Entropy_{Split}$	$\frac{5}{16}Entropy_{N1} + \frac{11}{16}Entropy_{N2} = \frac{5}{16} * 0.970 + \frac{11}{16} * 0.994 = 0.987$			
Δ_{info}	0.989-0.987=0.002			

Can Fly	N1 (Yes=1)	N1 Probability of Class	N2 (No=0)	N2 Probability of Class
Class 1: Mammals	1	$p_1 = \frac{1}{4}$	6	$p_1 = \frac{6}{12}$
Class 2: Non-Mammals	3	$p_2 = \frac{3}{4}$	6	$p_2 = \frac{6}{12}$
$Entropy$		$-\frac{1}{4}log_2\frac{1}{4} - \frac{3}{4}log_2\frac{3}{4} = 0.811$		$-\frac{6}{12}log_2\frac{6}{12} - \frac{6}{12}log_2\frac{6}{12} =$
$Entropy_{Split}$	$\frac{4}{16}Entropy_{N1} + \frac{12}{16}Entropy_{N2} = \frac{4}{16} * 0.811 + \frac{12}{16} * 1 = 0.953$			
Δ_{info}	0.989-0.953=0.036			

The feature "can fly" has higher information gain, it will be chose as the main first node.

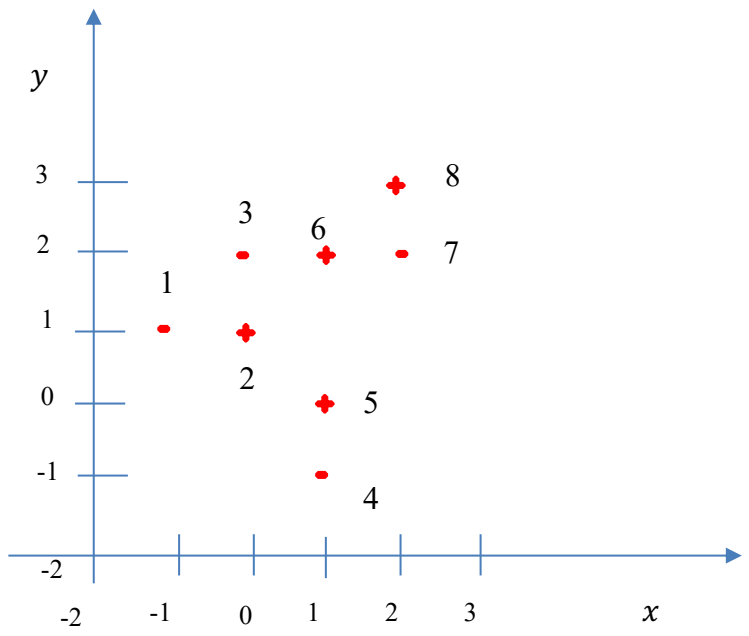
Question 4 (20 marks)

Suppose you have the following instance data where x and y are the 2 input features and the class label L

Instance	x	y	Class label L
1	-1	1	-
2	0	1	+
3	0	2	-
4	1	-1	-
5	1	0	+
6	1	2	+
7	2	2	-
8	2	3	+

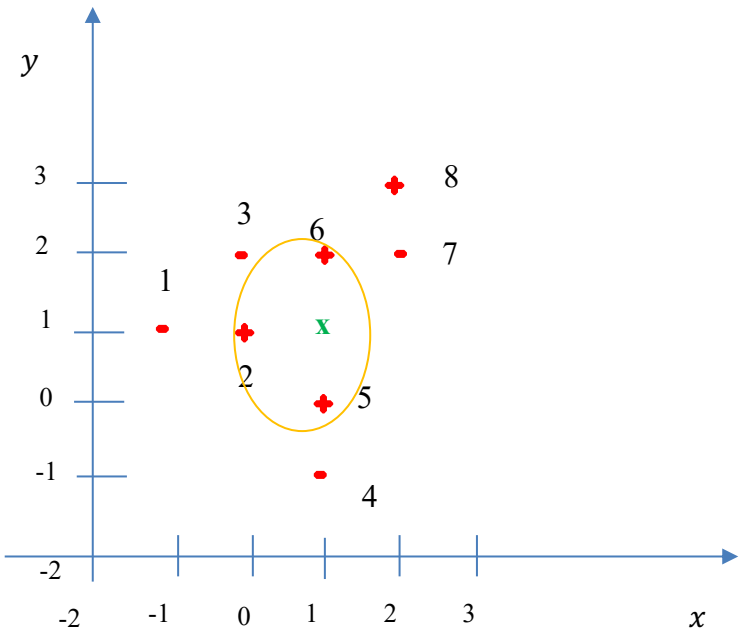
a) Draw the scatterplot in 2D space

<u>Solution :</u>



b) Suppose you want to predict the class of a new data point at $x = 1, y = 1$ using Euclidean distance in KNN where $K = 3$. In which class this data point belong to. Prove your working

Solution :



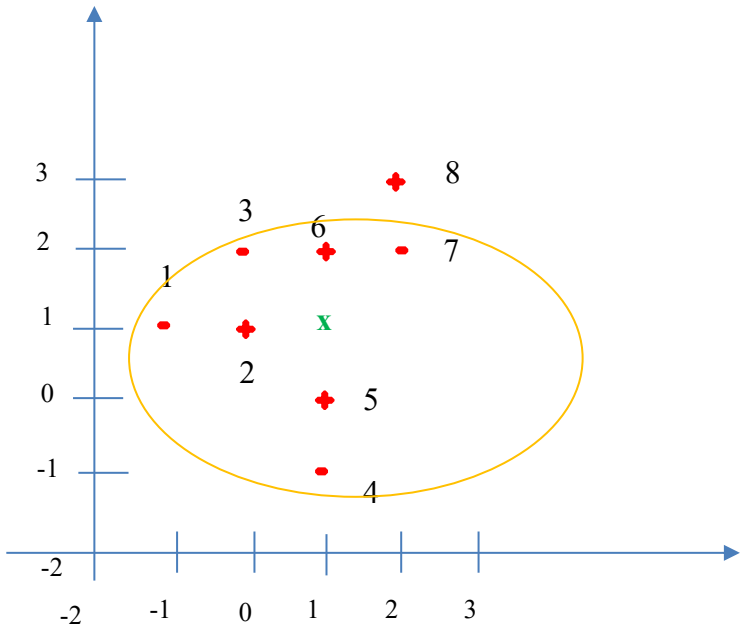
By visual inspection the nearest 3 neighbour points with new point $N: x = 1, y = 1$ are point 2, 5 and 6 and the Euclidean distance between them are

$$d(N,2) = \sqrt{(0-1)^2 + (1-1)^2} = 1$$
$$d(N,5) = \sqrt{(1-1)^2 + (0-1)^2} = 1$$
$$d(N,6) = \sqrt{(1-1)^2 + (2-1)^2} = 1$$

Since majority or all the point 2, 5 and 6 are class '+', therefore the new data point $N: x = 1, y = 1$ will be class '+'

c) If we use KNN with $k = 7$, which of the class does the data point $x = 1, y = 1$ belong to. Prove your working

Solution :



The Euclidean distance of the new point $N: x = 1, y = 1$ and all points are

$$d(N,1) = \sqrt{(-1-1)^2 + (1-1)^2} = 2$$
$$d(N,2) = \sqrt{(0-1)^2 + (1-1)^2} = 1$$
$$d(N,3) = \sqrt{(0-1)^2 + (2-1)^2} = 1.414$$
$$d(N,4) = \sqrt{(1-1)^2 + (-1-1)^2} = 2$$
$$d(N,5) = \sqrt{(1-1)^2 + (0-1)^2} = 1$$
$$d(N,6) = \sqrt{(1-1)^2 + (2-1)^2} = 1$$
$$d(N,7) = \sqrt{(2-1)^2 + (2-1)^2} = 1.414$$
$$d(N,8) = \sqrt{(2-1)^2 + (3-1)^2} = 2.236$$

Therefore the 7 nearest neighbours are points are all points except point 8 . There are 4 class '-' and 3 class '+'. Therefore the new point $N: x = 1, y = 1$ will be class '-'

---END OF PAPER---